Trainings

Select Service Tools

Recommended Cummins® Service Tools

- Cylinder Block Counterbore Cutting Kit, Part Number 5299988
- Depth gauge assembly, Part Number 3164438 or 5299194
- Counterbore ledge cutter drive unit, Part Number 3163785
- Chip removing unit, Part Number 3823461
- Crack Detection Kit, Part Number 3375432, or equivalent
- Counterbore Cutting Improvement Kit, Part Number 5395552.

Additional Service Items

- No additional service items required.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html)
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-037.html)
- Use the following procedure for the C Series engines. Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/41/41-002-004-tr.html)
- Use the following procedure for the C8.3G, C Gas Plus, and L Gas Plus engines. Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/39/39-002-004-tr.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-025.html)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-035.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/100/100-001-089.html)
- Use the following procedure for the C8.3G, C Gas Plus, and L Gas Plus engines. Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/39/39-001-046.html)
- Use the following procedure for the C Series engines. Remove the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/41/41-001-054.html)
- Use the following procedure for the C8.3G, C Gas Plus, and L Gas Plus engines. Remove the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/39/39-001-054-tr.html)

Initial Check

This repair procedure describes the method for machining the cylinder liner counterbore ledge using the C and L Series Cylinder Block Counterbore Cutting Kit, Part Number 5299988. The tool is designed to be used for light clean-up cuts.

Note : A different Cylinder Block Counterbore Cutting Kit exists for X series engines which is not compatible with C and L series engines. Be sure the correct kit is available before beginning the cutting procedure.

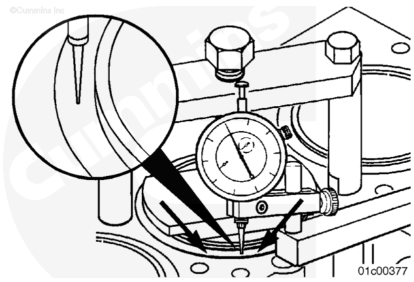
This procedure can be performed in or out of chassis.

The following Shims are available for C and L series engines:

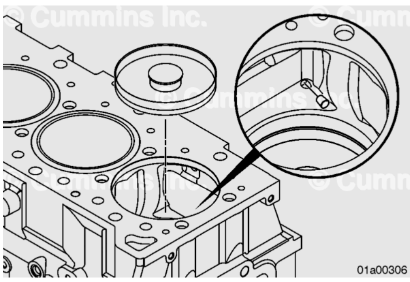
Part Number	Shim Thickness (mm)	Shim Thickness (in)
3924445	0.254	0.01
3924446	0.381	0.015
3924447	0.508	0.02
3924448	0.762	0.03
3924449	1	0.0394

Do **not** machine counterbore ledge to a depth greater than the thickest available shim.

Liner protrusion should be checked and verified. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/100/100-001-028.html)



Remove the cylinder liner of the cylinder being cut. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/100/100-001-028.html)



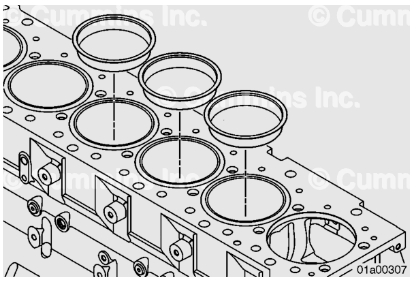
See the Cylinder Liner Counterbore Ledge Reuse Guidelines. Refer to Procedure 001-026 (/qs3/pubsys2/xml/en/procedures/39/39-001-026.html) in Section 1 to inspect the counterbore ledge surface areas for signs of pitting, fretting, or wear.

If fretting or wear is present on any cylinder, all six cylinders **must** be machined.

Machine

Cover the crankshaft, and plug any oil hole galleries and coolant passages to prevent machining chips from entering. Install cylinder bore plug, Part Number 5299991, into the cylinder bore.

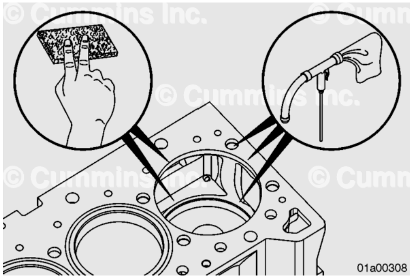
Install protective plug, Part Number 5299207, into the adjacent cylinders that are **not** being cut.



Make sure the top deck of the cylinder block is clean and free of burrs.

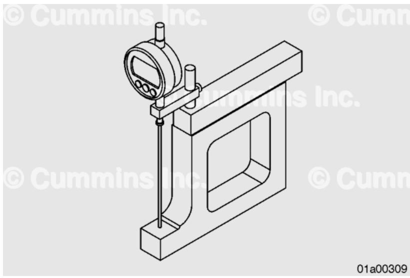
Use a medium grit stone, or equivalent, to clean the deck of the block and the counterbore ledge.

Vacuum and remove any debris from the deck of the block, counterbore ledge, and cylinder head capscrew holes.

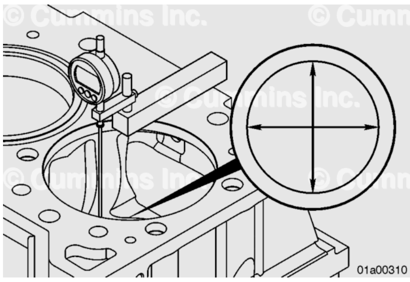


Zero the counterbore depth gauge assembly, Part Number 3164438 or 5299194, with the locating gauge, Part Number 5299989.

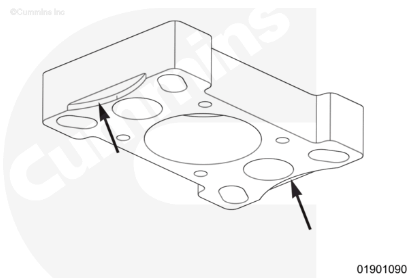
This will set the indicator to the original machining depth of 123 mm [4.843 in] from the cylinder deck to the counterbore ledge.



Measure the counterbore ledge in four locations in a crisscross pattern and mark the lowest point. Subtract the lowest number from the highest number. This is the minimum amount to be machined for clean-up to make the counterbore ledge even all the way around.



Use 30W non-detergent oil to lubricate the counterbore ledge cutter drive unit, Part Number 3163785. Fill the reservoir with 30W oil to maintain lubrication on the counterbore drive unit while machining the counterbore ledge. Assemble the mounting plate, Part Number 5299990, to the counterbore drive unit with the four hex key screws provided with the liner scallops facing down and tighten in a crisscross pattern.

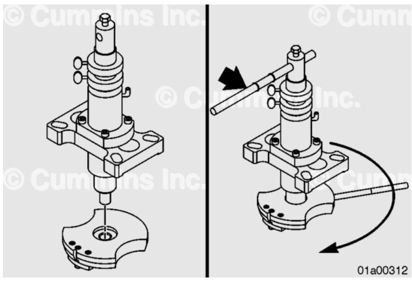


Torque Value: 27 n•m [239 in-lb]

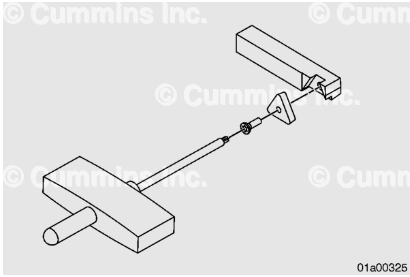
⚠ CAUTION ⚠

Make sure of proper installation of the counterbore cutter bit. Damage will occur if improperly installed.

Install the cutter plate, Part Number 5299993, to the counterbore drive unit. Use the cutter plate holder to tighten securely. The hole in the side of the cutter plate accepts the holder.

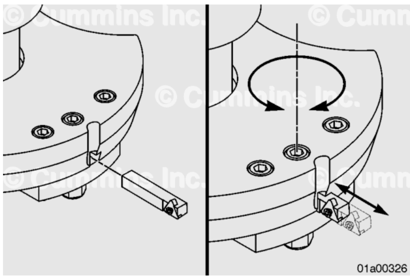


Position the cutter bit insert so the tapered edge faces the cutter bit. Install the cutter bit screw into the center of the cutter bit insert. The cutter bit insert is machined to allow the cutter bit screw to be fully recessed. Tighten the screw, using the provided cutter bit wrench, until the insert is fully seated against the cutter bit. Loosen the screw and tighten it again until there is no visible air gap around the insert and the bit.



Install the cutter bit, Part Number 5296531, into the cutter plate. The cutter bit must be installed with the part number facing to the left and the cutting edge facing to the left, as viewed from the bottom of the cutter plate. Make sure of proper installation of the cutter bit, as damage will occur if improperly installed.

Fully retract the cutter bit using the provided hex key wrench. Make sure the cutter bit can extend and retract freely.

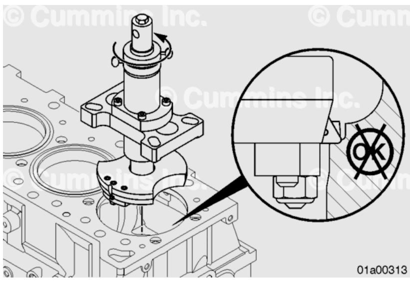


Make a paint pen mark on the counterbore ledge. This will indicate the machined area during the repair.

Position the counterbore ledge cutter unit into the cylinder bore by backing off the upper depth set collar and lower depth set collar in a **counterclockwise** direction.

This will allow the counterbore ledge cutting plate to lower and center itself on the counterbore ledge.

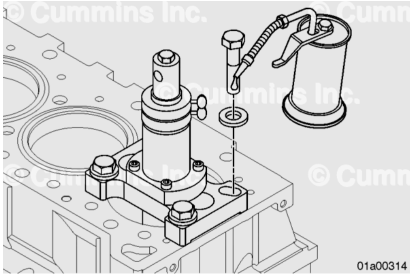
Care **must** be taken to verify that the base plate of the counterbore ledge cutting tool is positioned flat on the deck surface of the cylinder block, and the cutting plate is centered into the counterbore ledge with no side-to-side movement.



Secure the base plate to the cylinder block with the capscrews, special washers, and spacers included with the tool. Capscrews need to be lubricated with engine oil, installed hand-tight, and then tightened in a crisscross pattern.

Torque Value: 67 n•m [49 ft-lb]

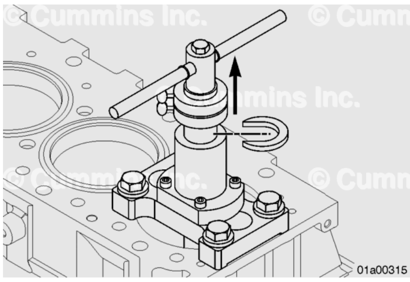
Install the T-handle onto the counterbore ledge cutter unit and adjust the lower stop collar in a **clockwise** rotation 0.127 mm [0.005 in] (each graduation on the collar is 0.025 mm [0.001 in]) to raise the cutter plate off the counterbore ledge and rotate the tool **clockwise**. Tighten the lower stop collar thumbscrew after every adjustment. No contact should be made with the counterbore ledge.



Note : If adequate clearance is not available, a 1/2 inch drive ratchet and 15/16 inch socket can be used in place of the T-handle for counterbore drive unit rotation.

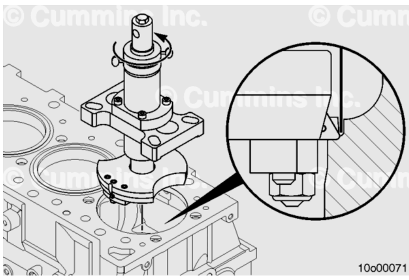
Raise the counterbore ledge cutter by grasping the T-handle and pulling in an upward motion.

Install the depth set collar (C-spacer) below the lower depth set collar and lower the tool until it makes contact with the depth set collar.



⚠ CAUTION ⚠

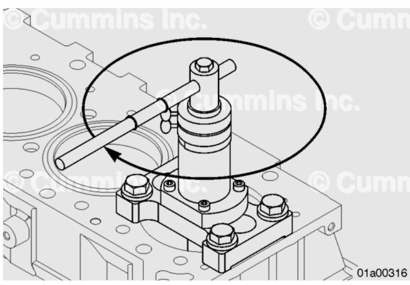
Make sure the cutter bit does not contact the step machined into the counterbore ledge. Damage to the cylinder block will occur during machining if the cutter bit is contacting the step.



Install a 0.812 mm [0.032 in] feeler gauge between the counterbore ledge wall and the cutter bit. This will make sure that the cutter bit does **not** make contact with the step machined into the counterbore ledge.

Extend the cutter bit with the hex key wrench installed in the friction screw until the cutter bit makes contact with the feeler gauge. Lock the bit down by tightening the two cutter bit hold-down capscrews with the hex key wrench.

Note : If adequate clearance is not available, a 1/2 inch drive ratchet and 15/16 inch socket can be used in place of the T-handle for counterbore drive unit rotation. The ratchet handle should initially be positioned so that it is directly above the cutter bit or as closely behind it as possible.



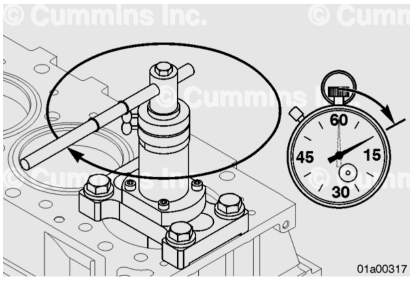
Rotate the counterbore ledge cutter unit **clockwise** for 360 degrees. A slight noise may be heard.

Lower the counterbore ledge cutter unit by adjusting the lower depth set collar in a **counterclockwise** rotation, until the cutter bit makes contact with the counterbore ledge. Raise the counterbore cutter unit 0.127 mm [0.005in] by adjusting the lower depth set collar in a **clockwise** direction.

Tighten the lower depth set collar thumbscrew. Rotate the counterbore ledge cutter unit **clockwise** for 360 degrees. No noise should be heard.

Loosen the two cutter bit hold down capscrews with the hex key wrench and extend the cutter bit until it makes contact with the counterbore ledge wall. Lock the cutter bit.

Lower the counterbore ledge cutter unit by 0.025 mm [0.001 in] at a time, while rotating the unit 360 degrees **clockwise** by adjusting the lower depth set collar in a **counterclockwise** rotation until it makes contact with the counterbore ledge.

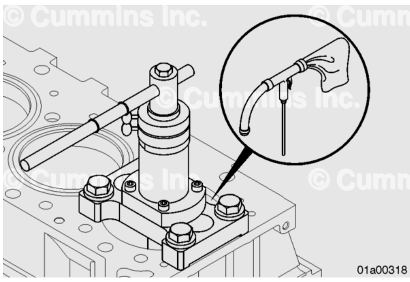


With the cutter bit making full contact with the counterbore ledge, rotate the counterbore ledge cutter unit 360 degrees **clockwise** to clean up the counterbore ledge. Observe the change made to the paint pen mark to confirm correct cutter bit extension.

Note : If adequate clearance is not available, a 1/2 inch drive ratchet and 15/16 inch socket can be used in place of the T-handle for counterbore drive unit rotation. The ratchet handle should initially be positioned so that it is directly above the cutter bit or as closely behind it as possible.

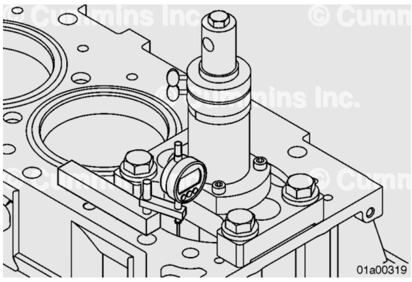
Note : Care must be taken to maintain consistent downward pressure on the top of the T-handle or ratchet head during the cutting portion of the stroke. Each 360 degree revolution should take at least 10 seconds. Any deviation in speed will result in chatter on the counterbore ledge.

Remove any chips on the counterbore ledge with a chip removing unit, Part Number 3823461.



With the counterbore depth gauge assembly, Part Number 3164438 or 5299194,, measure the depth of the counterbore ledge. This will determine what the final cut depth will be.

Note : Counterbore ledge machining should not exceed the thickness of the largest available shim.



Adjust the upper depth set collar **counterclockwise** to the final depth cut.

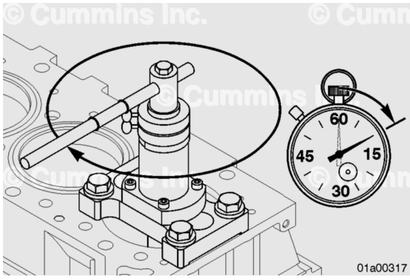
Example: If cutting 0.127 mm [0.005 in], adjust the upper depth set collar to contact the lower depth step collar and then back it off 0.127 mm [0.005 in].

Adjust the lower depth set collar **counterclockwise** 0.025 mm [0.001 in] at a time until the final cut is reached. Tighten the thumbscrew after each adjustment. A cut is complete when the lower depth set collar is firmly contacting the c-ring.

Rotate the counterbore ledge cutter unit **clockwise** with the T-handle positioned so the long portion of the handle is directly above the cutter bit.

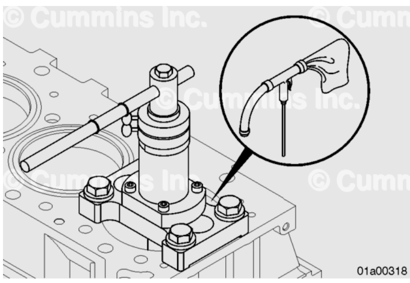
Note : If adequate clearance is not available, a 1/2 inch drive ratchet and 15/16 inch socket can be used in place of the T-handle for counterbore drive unit rotation. The ratchet handle should initially be positioned so that it is directly above the cutter bit or as closely behind it as possible.

Note : Care must be taken to maintain consistent downward pressure on the top of the T-handle or ratchet head during the cutting portion of the stroke. Each 360 degree revolution should take at least 10 seconds. Any deviation in speed will result in chatter on the counterbore ledge.

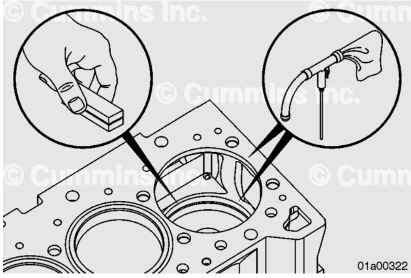


The counterbore ledge **must** be vacuumed out before each measurement.

Note : It is recommend stopping with 0.025 mm [0.001 in] left to cut and take one final measurement to verify the cutting depth.



Adjust the lower depth set collar by rotating it **clockwise** to raise the cutter plate off the counterbore ledge.

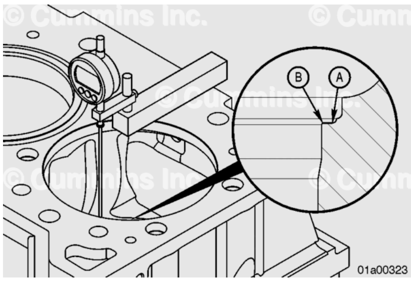


Loosen the two cutter bit hold-down capscrews with the hex key wrench and retract the cutter bit into the cutter plate.

Remove the capscrews that secured the counterbore ledge cutter unit to the deck of the cylinder block and remove the tool from the engine.

Clean any machined chips from the counterbore chip catcher, counterbore ledge, and the deck of the cylinder block. Use a medium grit stone, or equivalent, to clean up the leading edge of the counterbore ledge.

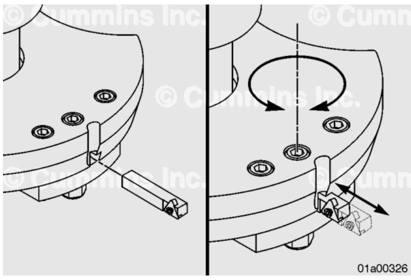
With the counterbore depth gauge assembly, Part Number 3164438 or 5299194, measure the depth of the counterbore ledge and the ledge angle in four places.



The four measurements **must not** vary more than 0.025 mm [0.001 in]. If the measurements exceed the specifications, the counterbore ledge **must** be machined again.

Dimension A should be 0.0127 mm [0.0005 in] lower or equal to Dimension B.

Remove the chip catcher and any oil gallery or coolant passage plugs that were installed.



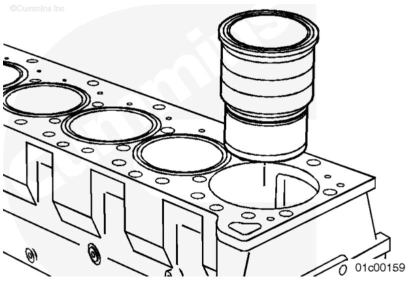
Inspect the counterbore seating area for cracking using Crack Detection Kit, Part Number 3375432, or equivalent. If there are any cracks present, the block counterbore ledges may be machined to the next shim thickness, if a thicker shim is offered.

Note : Do not remove more material from the counterbore ledge than can be made up with the present shims offered.

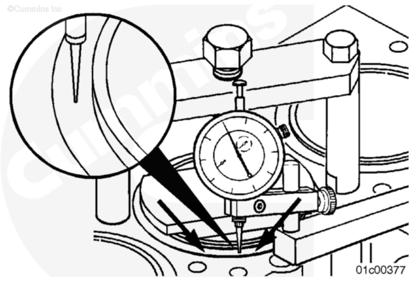
If the counterbore ledge machining is repeated, recheck the seating area for cracking, using Crack Detection Kit, Part Number 3375432, or equivalent. If cracks are still present, the cylinder block will need to be replaced.

Install a hardened bronze shim, Part Number 3924445, 3924446, 3924447, 3924448, or 3924449, into the cylinder that was cut. Do **not** stack shims.

Install a new cylinder liner into the cylinder that was cut. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/100/100-001-028.html)



Liner protrusion needs to be checked and verified. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/100/100-001-028.html)



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Use the following procedure for the C Series engines. Install the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/41/41-001-054.html)
- Use the following procedure for the C8.3G, C Gas Plus, and L Gas Plus engines. Remove the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/39/39-001-054-tr.html)
- Use the following procedure for the C8.3G, C Gas Plus, and L Gas Plus engines. Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/39/39-001-046.html)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/100/100-001-089.html)

- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-035.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-025.html)
- Use the following procedure for the C Series engines. Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/39/39-002-004-tr.html)
- Use the following procedure for the C8.3G, C Gas Plus, and L Gas Plus engines. Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/41/41-002-004-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/100/100-007-037.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.

Last Modified: 22-Jan-2021